

L1.9 L'Hôpital's Rule

§6.8 — what you need to know

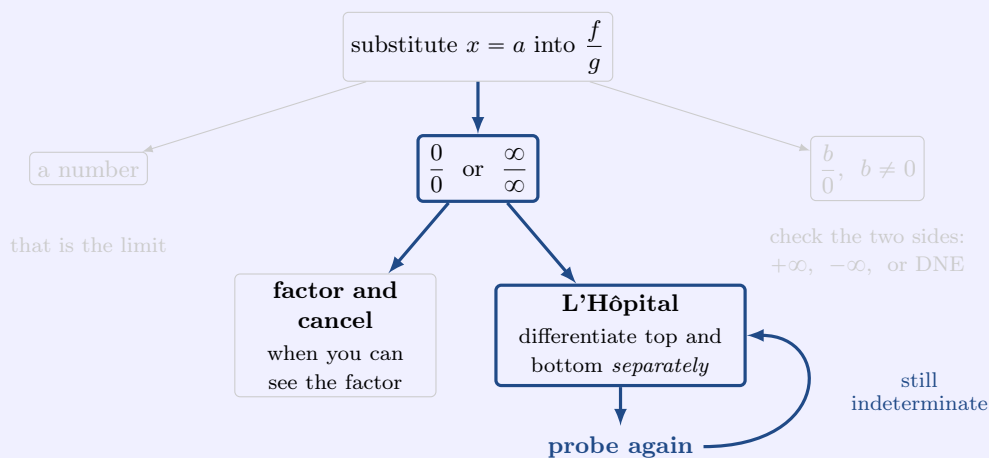
L'HÔPITAL'S RULE

$$\text{if } \frac{f(a)}{g(a)} = \frac{0}{0} \text{ or } \frac{\infty}{\infty}, \quad \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

“when both hit zero together, compare how fast they leave”

Both are zero at a , so each function's **value** just past a is its **change**: $f(a + dx) = 0 + \Delta f$. The quotient is therefore $\Delta f / \Delta g$, and replacing each change by its differential gives $\frac{f'(a) dx}{g'(a) dx}$. **The dx divides out** — which is the cancel step from L1.2, done with derivatives, on the factor algebra could not reach.

Try $f(a)$ first — the chart, with one new branch



Nothing changed on the old branches. $\frac{0}{0}$ simply has a second exit now — and $\frac{0}{0}$ still means *no information yet*, never *no limit*.

APPLYING THE RULE

① **Probe.** Substitute. Only $\frac{0}{0}$ and $\frac{\infty}{\infty}$ open the door.

② **Differentiate top and bottom separately.**

$$\frac{f}{g} \longrightarrow \frac{f'}{g'}$$

③ **Probe again.** Still indeterminate, repeat. Otherwise **stop** — you are done.

Worked. $\lim_{x \rightarrow \infty} \frac{3x^2 - 4x + 1}{5x^2 - 2x + 1}$

$$\frac{\infty}{\infty} \rightarrow \lim_{x \rightarrow \infty} \frac{6x - 4}{10x - 2} \rightarrow \frac{\infty}{\infty}$$

$$\rightarrow \lim_{x \rightarrow \infty} \frac{6}{10} = \frac{3}{5}$$

Two applications, and the answer is the ratio of the leading coefficients — the rule agrees with what you already knew.

WHAT GOES WRONG

~~✗ it is the quotient rule~~ → **differentiate the top and the bottom separately — nothing is subtracted, nothing is squared**

~~✗ keep differentiating until it looks clean~~ → **re-probe after every application; once the form is a number you are finished**

~~✗ any quotient with 0 underneath~~ → $\frac{b}{0}$ with $b \neq 0$ is not indeterminate — that is the **infinite-limit case**

~~✗ $\frac{0}{0}$ means the limit does not exist~~ → it means *no information yet*: $\frac{5x}{x} \rightarrow 5$, $\frac{x^2}{x} \rightarrow 0$, all the same form

~~✗ reach for the rule as soon as you see a fraction~~ → **probe first — if substituting gives a number, that number *is* the limit**

Four more indeterminate forms exist — $0 \cdot \infty$, $\infty - \infty$, 0^0 , ∞^0 and 1^∞ . None of them is a ratio, so none of them is ready for this rule; each has to be rewritten as one first. That is Chapter 6.